	= 1.7 m	[1] answer
2(a)	As the car speeds up, the resistive forces acting on it will increase as drag force is proportional to square of velocity. Resultant force is equal to thrust minus resistive force. Since thrust is constant while resistive force increase at a decreasing rate, resultant force will decrease at a decreasing rate	[1] drag proportional to square of velocity(high speed) [1] resultant decrease at decreasing rate (no marks for decrease)
2(b)	At t = 0, drag force is zero as car is at rest. Therefore only force providing resultant force is the thrust of the rocket. By Newton's 2nd law $F = \frac{d(mv)}{dt} = \frac{vd(m)}{dt}$ $5000 = v(1.3)$ $v = 3.8 \times 10^3 \text{ m s}^{-1}$	[1] deducing that drag force is zero [1] correct substitution [1] correct answer
2(c)	Impulse = area under the graph of resultant force vs time 4 trapeziums + 1 triangle = [½ × 2.5 × (5000+2800)] + [½ × 2.5 × (2800+ 1600)] + [½ × 2.5 × (1600+800)] + [½ × 2.5 × (800+400)] + [½ × 2.5 × 400] = 20250 kg m s^{-1}	[1] evidence to show area under graph [1] method [1] accuracy between 19800-22000
2(d)	total mass = 600 – (1.3 x 15.0) = 580 kg	[1]

3(a)	The first law of thermodynamics states that the increase in	[2] or zero